



## **GOSFORD HIGH SCHOOL**

**September 2019**

### **Preliminary Mathematics Extension 1**

#### **Assessment Task 3**

##### **General Instructions**

- Reading time – 5 minutes
- Working time – 1 hour 30 minutes
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided
- All necessary working should be shown in Questions 9 - 25

**Total Marks – 55**

##### **Section I - 8 marks**

- Attempt Questions 1 - 8
- Allow about 12 minutes for this section

##### **Section II - 47 marks**

- Attempt Questions 9 - 15
- Allow about 1 hour 18 minutes for this section

STUDENT NUMBER/NAME: .....

## Section I

**8 Marks**

**Attempt Questions 1-8**

**Allow about 12 minutes for this section.**

Select the alternative A, B, C or D that best answers the question and indicate your choice with a cross (X) in the appropriate space on the grid below.

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|          | A | B | C | D |
|----------|---|---|---|---|
| <b>1</b> |   |   |   |   |
| <b>2</b> |   |   |   |   |
| <b>3</b> |   |   |   |   |
| <b>4</b> |   |   |   |   |
| <b>5</b> |   |   |   |   |
| <b>6</b> |   |   |   |   |
| <b>7</b> |   |   |   |   |
| <b>8</b> |   |   |   |   |

## Section I

## 8 Marks

## Attempt Questions 1-8

Allow about 12 minutes for this section.

Use the multiple-choice answer sheet for questions 1-8

- 
1. What is the Cartesian equation of the line with parametric equations  $x = t + 2$  and  $y = 2t + 1$  ? 1
- (A)  $2x + y + 3 = 0$   
(B)  $2x - y - 3 = 0$   
(C)  $2x + y - 3 = 0$   
(D)  $2x - y + 3 = 0$
2. What is the domain of the function  $f(x) = \sqrt{x^2 - 2x}$  ? 1
- (A)  $x \geq 2$   
(B)  $x > 2$   
(C)  $x \leq 0$  or  $x \geq 2$   
(D)  $x < 0$  or  $x > 2$
3. For what values of  $k$  does the polynomial  $P(x) = (k^3 - k)x^4 + (k^2 - k)x^3 + x^2$  have degree 3? 1
- (A)  $k = -1$   
(B)  $k = -1$  or  $k = 0$   
(C)  $k = -1$  or  $k = 1$   
(D)  $k = -1, k = 0$  or  $k = 1$
4. What is the remainder when  $P(x) = x^2 - (k+1)x - (k+1)^2$  is divided by  $(x+k)$  ? 1
- (A)  $k^2 + k + 1$   
(B)  $k^2 - k - 1$   
(C)  $k^2 + k - 1$   
(D)  $k^2 - k + 1$
5. What is the value of  $\cos^{-1}(\cos \frac{5\pi}{6}) - \sin^{-1}(\sin \frac{5\pi}{6})$  ? 1
- (A) 0  
(B)  $\frac{\pi}{3}$   
(C)  $-\frac{\pi}{2}$   
(D)  $\frac{2\pi}{3}$

**Marks**  
**1**6. Which of these is NOT an expression for  $\tan 2x$  ?

- (A)  $\frac{\sin 4x}{1 + \cos 4x}$
- (B)  $\frac{1 - \cos 4x}{\sin 4x}$
- (C)  $\frac{1}{1 - \tan x} - \frac{1}{1 + \tan x}$
- (D)  $\frac{\tan x}{1 - \tan x} - \frac{\tan x}{1 + \tan x}$

7. A group consists of 3 pairs of brothers and sisters from 3 different families. In how many ways can the group be seated in a circle so that the members of the same family are sitting opposite each other?

**1**

- (A) 8
- (B) 12
- (C) 16
- (D) 20

8. Which of the following is obtained by expanding  $(2 + \sqrt{3})^3$  ?**1**

- (A)  $8 + 3\sqrt{3}$
- (B)  $14 + 7\sqrt{3}$
- (C)  $20 + 11\sqrt{3}$
- (D)  $26 + 15\sqrt{3}$

**47 Marks****Marks****Attempt Questions 9-25****Allow about 1 hour and 18 minutes for this section.**

Answer the questions in the spaces provided. Sufficient space is provided for typical responses.

Your responses should include relevant mathematical reasoning and/or calculations.

Extra writing space is provided at the back of this booklet.

If you use this space, clearly indicate which question you are answering

**Question 9 (1 mark)**Solve the inequality  $|2x - 1| \leq 3$ .**1**

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**Question 10 (3 marks)**Find the coefficient of the term containing  $x^3$  in the expansion of  $\left(2x + \frac{3}{x}\right)^{11}$ .**3**

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**Question 11 (2 marks)**Show that  $\frac{\cos^2 x - \cos 2x}{\sin^2 x + \cos 2x} = \tan^2 x$ .**2**

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**Question 12 (3 marks)**Solve the inequality  $\frac{x^2-2}{x} < 1$ .

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**Question 13 (3 marks)**Consider the polynomial  $P(x) = x^3 - px^2 + qx - pq$ , where  $q > 0$ .(a) Show that  $(x - p)$  is a factor of  $P(x)$ .

1

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(b) Show that the equation  $P(x) = 0$  has only one real root.

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**Question 14 (3 marks)**

**Marks**

Solve the equation  $\sin 2x = \sin x$  for  $0 \leq x \leq 2\pi$ .

**3**

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**Question 15 (3 marks)**

Find the number of ways that a team of 4 can be chosen from 5 boys and 6 girls

(a) with equal numbers of boys and girls.

**1**

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(b) with a majority of girls.

**2**

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**Question 16 (3 marks)**

The equation  $x^3 + 6x^2 + 4x + 2 = 0$  has roots  $\alpha$ ,  $\beta$  and  $\gamma$ .

(a) Find the value of  $\frac{\alpha\beta}{2} + \frac{\beta\gamma}{2} + \frac{\gamma\alpha}{2}$ .

**1**

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(b) Find the value of  $\frac{2}{\alpha\beta} + \frac{2}{\beta\gamma} + \frac{2}{\gamma\alpha}$ .

**2**

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**Question 17 (3 marks)**

- (a) Find the domain and range of the function  $f(x) = 2\cos^{-1}(x-1)$ . 2

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- (b) Sketch the graph of the function  $f(x) = 2\cos^{-1}(x-1)$  showing the intercepts on the axes and the coordinates of the endpoints. 1

**Question 18 (3 marks)**

- (a) Write down the binomial expansion of  $(1+x)^n$ . 1

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- (b) Hence show that  ${}^nC_1 + {}^nC_2 + {}^nC_3 + \dots + {}^nC_n = 2^n - 1$ . 2

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Student name / number \_\_\_\_\_

Marks

**Question 19 (2 marks)**

Find the multiplicity of the root 1 of the equation  $P(x) = 0$  where

2

$$P(x) = x^4 - 5x^3 + 9x^2 - 7x + 2.$$

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**Question 20 (2 marks)**

Sketch the graph of the function  $f(x) = \tan^{-1} x$  showing the intercepts on the axes and the equations of the asymptotes.

2

**Question 21 (4 marks)**

- (a) Find in fully expanded form the Cartesian equation of the circle with parametric equations 2

$$x = 4 + 5\cos\theta \quad \text{and} \quad y = 3 + 5\sin\theta.$$

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- (b) Sketch the graph of this circle showing the intercepts on the axes. 2

**Marks**

**Question 22 (4 marks)**

Consider the function  $f(x) = x^2 - 2$ ,  $0 \leq x \leq 4$ .

- (a) Find the inverse function  $f^{-1}(x)$  and state its domain.

2

[illegible]

- (b) On the same diagram sketch the graphs of the function  $f(x)$  and the inverse function  $f^{-1}(x)$  showing the coordinates of the endpoints and the intercepts on the axes.

2

**Question 23 (2 marks)**

Find the number of ways in which the letters of the word HEXAGON can be arranged in a straight line

(a) so that the vowels are all next to each other.

1

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(b) so that no two consonants are next to each other

1

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**Question 24 (2 marks)**

If  $\cos^{-1} x - \sin^{-1} x = \frac{\pi}{4}$ , find the exact value of  $\cos^{-1} x$ .

2

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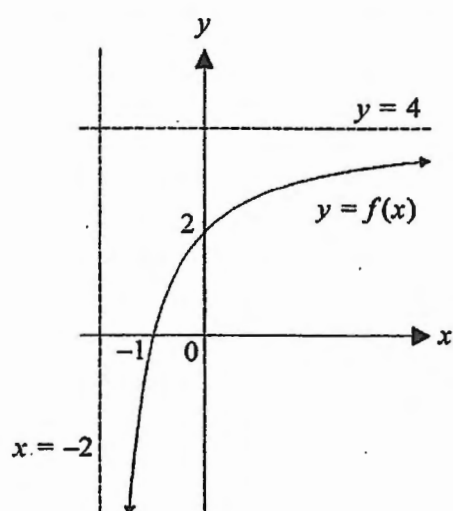
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**Question 25 (4 marks)**

The diagram shows the graph of the curve  $y = f(x)$  where  $f(x) = \frac{4x+4}{x+2}$ ,  $x > -2$ .

On separate diagrams sketch the graphs of the following curves, showing clearly the intercepts on the axes and the equations of any asymptotes.



(a)  $y = |f(x)|$

1

(b)  $y^2 = f(x)$

1

(c)  $y = \frac{1}{f(x)}$

2

Student name / number \_\_\_\_\_

### Marks

**Section II** Extra writing space.

**If you use this space, clearly indicate which question you are answering.**

[illegible]

Student name / number \_\_\_\_\_

### Marks

[illegible]



## **GOSFORD HIGH SCHOOL**

**September 2019**

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**Section I**

**8 Marks**

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|   | A | B | C | D |
|---|---|---|---|---|
| 1 |   | X |   |   |
| 2 |   |   | X |   |
| 3 | X |   |   |   |
| 4 |   | X |   |   |
| 5 |   |   |   | X |
| 6 |   |   |   | X |
| 7 | X |   |   |   |
| 8 |   |   |   | X |

## Section I

## 8 Marks

## Attempt Questions 1-8

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4. What is the remainder when  $P(x) = x^2 - (k+1)x - (k+1)^2$  is divided by  $(x+k)$  ? 1
  - (A)  $k^2 + k + 1$
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5. What is the value of  $\cos^{-1}\left(\cos \frac{5\pi}{6}\right) - \sin^{-1}\left(\sin \frac{5\pi}{6}\right)$  ? 1
  - (A) 0
  - (B)  $\frac{\pi}{3}$
  - (C)  $-\frac{\pi}{2}$
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**Marks**  
**1**6. Which of these is NOT an expression for  $\tan 2x$  ?

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(C)  $\frac{1}{1 - \tan x} - \frac{1}{1 + \tan x}$

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(B) 12

(C) 16

(D) 20

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**Question 9 (1 mark)**Solve the inequality  $|2x - 1| \leq 3$ .**1**

$$\begin{aligned} -3 &\leq 2x - 1 \leq 3 \\ -2 &\leq 2x \leq 4 \\ -1 &\leq x \leq 2 \end{aligned}$$

**Question 10 (2 marks)**Find the coefficient of the term containing  $x^3$  in the expansion of  $(2x + \frac{3}{x})^{11}$ .**3**

$$\begin{aligned} \text{Term containing } x^3 \text{ is } {}^{11}C_7 (2x)^7 \left(\frac{3}{x}\right)^4 \\ = {}^{11}C_7 \times 2^7 \times 3^4 \times x^{-4} \\ = {}^{11}C_7 \times 2^7 \times 3^4 \times x^3 \\ = 3421440 x^3 \end{aligned}$$

 $\therefore$  coefficient is 3421440**Question 11 (2 marks)**Show that  $\frac{\cos^2 x - \cos 2x}{\sin^2 x + \cos 2x} = \tan^2 x$ .**2**

$$\begin{aligned} \frac{\cos^2 x - \cos 2x}{\sin^2 x + \cos 2x} &= \frac{\cos^2 x - (\cos^2 x - \sin^2 x)}{\sin^2 x + (\cos^2 x - \sin^2 x)} \\ &= \frac{\sin^2 x}{\cos^2 x} \\ &= \tan^2 x \end{aligned}$$

**Question 12 (3 marks)**Solve the inequality  $\frac{x^2-2}{x} < 1$ .

3

$$\frac{x^2-2}{x} < 1$$

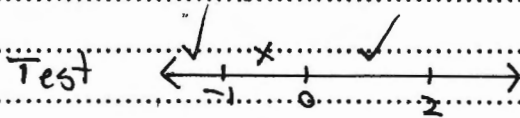
$$x \neq 0$$

$$x(x^2-2) < x^2$$

$$x(x^2-2)-x^2 < 0$$

$$x(x^2-x-2) < 0$$

$$x(x+1)(x-2) < 0$$



$$\therefore x < -1 \text{ or } 0 < x < 2$$

**Question 13 (3 marks)**Consider the polynomial  $P(x) = x^3 - px^2 + qx - pq$ , where  $q > 0$ .(a) Show that  $(x-p)$  is a factor of  $P(x)$ .

1

$$\begin{aligned} P(p) &= p^3 - p \cdot p^2 + qp - pq \\ &= 0 \end{aligned}$$

$\therefore x-p$  is a factor

(b) Show that the equation  $P(x) = 0$  has only one real root.

2

$$\begin{aligned} P(x) &= x^3 - px^2 + qx - pq \\ &= x^2(x-p) + q(x-p) \\ &= (x^2+q)(x-p) \end{aligned}$$

now  $x^2+q=0$  has no real roots $\therefore P(x)=0$  has only one real root  $p$

## Question 14 (3 marks)

Solve the equation  $\sin 2x = \sin x$  for  $0 \leq x \leq 2\pi$ .

3

$$\sin 2x = \sin x \quad 0 \leq x \leq 2\pi$$

$$2\sin x \cos x = \sin x$$

$$\sin x (2\cos x - 1) = 0$$

$$\sin x = 0 \text{ or } \cos x = \frac{1}{2}$$

$$x = 0, \pi, 2\pi \quad x = \frac{\pi}{3}, \frac{5\pi}{3}$$

$$\therefore x = 0, \frac{\pi}{3}, \pi, \frac{5\pi}{3}, 2\pi$$

## Question 15 (3 marks)

Find the number of ways that a team of 4 can be chosen from 5 boys and 6 girls

(a) with equal numbers of boys and girls.

1

$${}^5C_2 \times {}^6C_2 = 150$$

(b) with a majority of girls.

2

$${}^6C_4 \times {}^5C_0 + {}^6C_3 \times {}^5C_1 = 115$$

## Question 16 (3 marks)

The equation  $x^3 + 6x^2 + 4x + 2 = 0$  has roots  $\alpha$ ,  $\beta$  and  $\gamma$ .(a) Find the value of  $\frac{\alpha\beta}{2} + \frac{\beta\gamma}{2} + \frac{\gamma\alpha}{2}$ .

1

$$\alpha\beta + \beta\gamma + \gamma\alpha = 4$$

$$\therefore \frac{\alpha\beta}{2} + \frac{\beta\gamma}{2} + \frac{\gamma\alpha}{2} = 2$$

(b) Find the value of  $\frac{2}{\alpha\beta} + \frac{2}{\beta\gamma} + \frac{2}{\gamma\alpha}$ .

2

$$\frac{2}{\alpha\beta} + \frac{2}{\beta\gamma} + \frac{2}{\gamma\alpha} = \frac{2(\gamma + \alpha + \beta)}{\alpha\beta\gamma}$$

$$= \frac{2 \times (-6)}{-2}$$

$$= 6$$

## Marks

## Question 17 (3 marks)

- (a) Find the domain and range of the function
- $f(x) = 2\cos^{-1}(x-1)$
- .

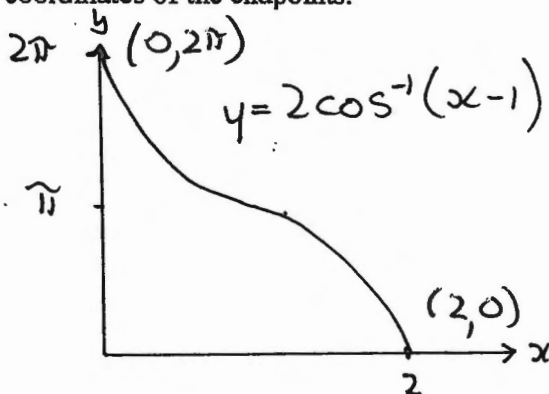
2

$$\text{Domain: } -1 \leq x-1 \leq 1 \quad \text{Range: } 0 \leq \cos^{-1}(x-1) \leq \pi$$

$$\therefore 0 \leq x \leq 2 \quad \therefore 0 \leq f(x) \leq 2\pi$$

- (b) Sketch the graph of the function
- $f(x) = 2\cos^{-1}(x-1)$
- showing the intercepts on the axes and the coordinates of the endpoints.

1



## Question 18 (3 marks)

- (a) Write down the binomial expansion of
- $(1+x)^n$
- .

1

$$(1+x)^n = {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + {}^nC_3 x^3 + \dots + {}^nC_n x^n$$

- (b) Hence show that
- ${}^nC_1 + {}^nC_2 + {}^nC_3 + \dots + {}^nC_n = 2^n - 1$
- .

2

substituting  $x=1$ ,  $2^n = {}^nC_0 + {}^nC_1 + {}^nC_2 + {}^nC_3 + \dots + {}^nC_n$

now  ${}^nC_0 = 1 \therefore 2^n - 1 = {}^nC_1 + {}^nC_2 + {}^nC_3 + \dots + {}^nC_n$

**Question 19 (2 marks)**Find the multiplicity of the root 1 of the equation  $P(x)=0$  where

2

$$P(x) = x^4 - 5x^3 + 9x^2 - 7x + 2$$

$$P(x) = x^4 - 5x^3 + 9x^2 - 7x + 2 \quad P(1) = 0$$

$$P'(x) = 4x^3 - 15x^2 + 18x - 7 \quad P'(1) = 0$$

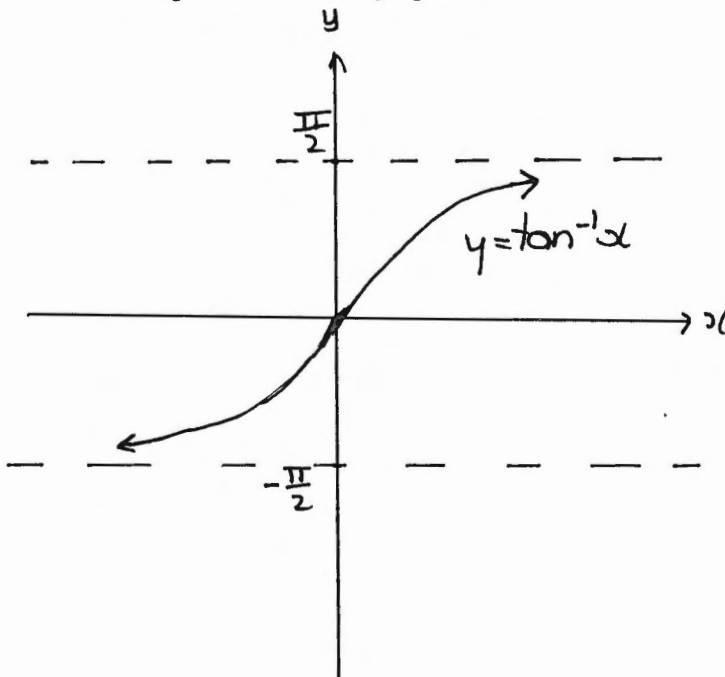
$$P''(x) = 12x^2 - 30x + 18 \quad P''(1) = 0$$

$$P'''(x) = 24x - 30 \quad P'''(1) \neq 0$$

$\therefore 1$  is a root of multiplicity of 3

**Question 20 (2 marks)**Sketch the graph of the function  $f(x) = \tan^{-1} x$  showing the intercepts on the axes and the equations of the asymptotes.

2



**Question 21 (4 marks)**

(a) Find in fully expanded form the Cartesian equation of the circle with parametric equations

2

$$x = 4 + 5\cos\theta \text{ and } y = 3 + 5\sin\theta.$$

$$x = 4 + 5\cos\theta \quad y = 3 + 5\sin\theta$$

$$\cos\theta = \frac{x-4}{5}$$

$$\sin\theta = \frac{y-3}{5}$$

now

$$\cos^2\theta + \sin^2\theta = 1$$

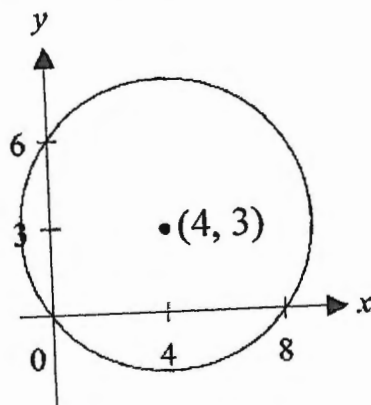
$$\left(\frac{x-4}{5}\right)^2 + \left(\frac{y-3}{5}\right)^2 = 1$$

$$(x-4)^2 + (y-3)^2 = 25$$

$$x^2 - 8x + 4^2 + y^2 - 6y + 3^2 = 25$$

(b) Sketch the graph of this circle showing the intercepts on the axes.

2



**Question 22 (4 marks)**

Consider the function  $f(x) = x^2 - 2$ ,  $0 \leq x \leq 4$ .

- (a) Find the inverse function  $f^{-1}(x)$  and state its domain.

2

$$y = x^2 - 2, \quad 0 \leq x \leq 4$$

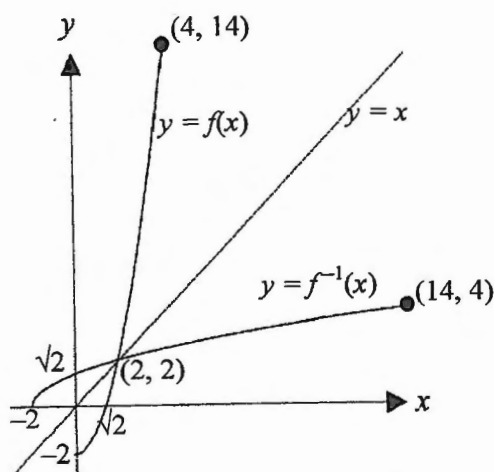
$$x^2 = y + 2$$

$$x = \sqrt{y+2}, \quad -2 \leq y \leq 14$$

$$f^{-1}(x) = \sqrt{x+2}, \quad -2 \leq x \leq 14$$

- (b) On the same diagram sketch the graphs of the function  $f(x)$  and the inverse function  $f^{-1}(x)$  showing the coordinates of the endpoints and the intercepts on the axes.

2



**Question 23 (2 marks)**

Find the number of ways in which the letters of the word HEXAGON can be arranged in a straight line

(a) so that the vowels are all next to each other. 1

(EAO) = vowels H X G N can be arranged in  $5!$  ways.  
 (EAO) can be arranged in  $3!$  ways.  
 $5! \times 3! = 720$  ways

(b) so that no two consonants are next to each other 1

V = vowels C = consonants  
 can only be arranged as C V C V C V C V C  
 C arrangements =  $4!$  C arrangements =  $3!$   
 $4! \times 3! = 144$  ways

**Question 24 (2 marks)**

If  $\cos^{-1} x - \sin^{-1} x = \frac{\pi}{4}$ , find the exact value of  $\cos^{-1} x$ . 2

$$\cos^{-1} x - \sin^{-1} x = \frac{\pi}{4} \quad (1)$$

now  $\cos^{-1} x + \sin^{-1} x = \frac{\pi}{2} \quad (2)$

(1) + (2)

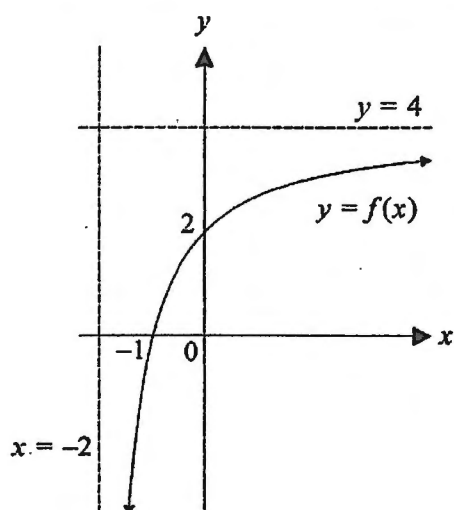
$$2\cos^{-1} x = \frac{3\pi}{4}$$

$$\cos^{-1} x = \frac{3\pi}{8}$$

**Question 25 (4 marks)**

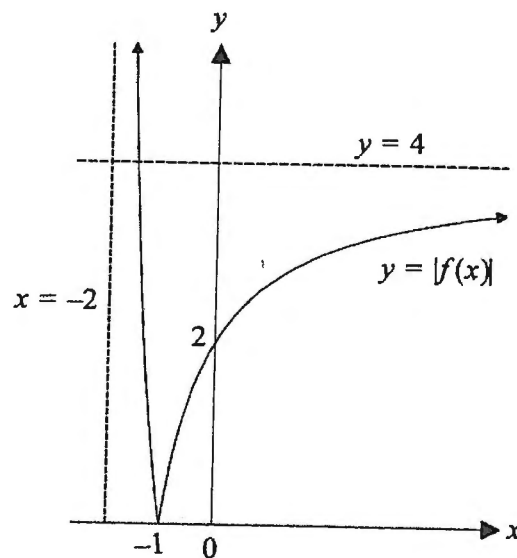
The diagram shows the graph of the curve  $y = f(x)$  where  $f(x) = \frac{4x+4}{x+2}$ ,  $x > -2$ .

On separate diagrams sketch the graphs of the following curves, showing clearly the intercepts on the axes and the equations of any asymptotes.



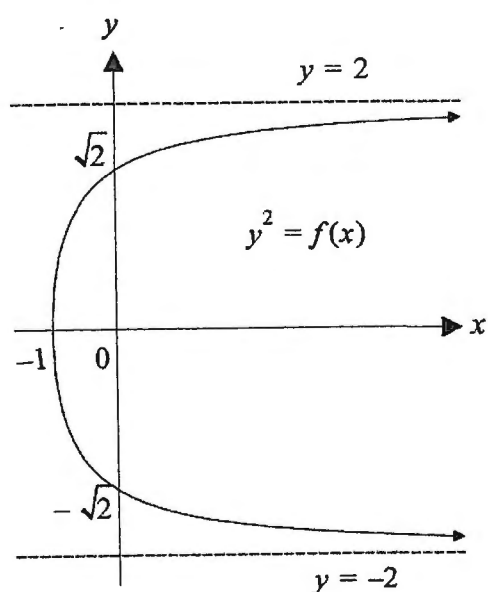
(a)  $y = |f(x)|$

1



(b)  $y^2 = f(x)$

1



(c)  $y = \frac{1}{f(x)}$

2

